MILITARY SPECIFICATION SHEET

ELECTRON TUBE, CATHODE RAY

TYPE 3KP1

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

DESCRIPTION: Electrostatic deflection and focus

DIMENSIONS AND PIN CONNECTIONS: See figure 1

ABSOLUTE RATINGS:

Parameter:	Ef	Ec1	ed	Eb1	Eb2	Light output	Rg	Zd	Ehk	Alt
Unit:	V	Vdc	v	Vdc	Vdc	ftL	Meg	Meg	Vdc	ft
Maximum:	6.9	-200	550	1,100	2,750	---	1.5	1.0	±125	10,000
Minimum:	5.7	0	---	---	550 (Note 2)	4.0	---	---	---	---
TEST CONDITIONS:	6.3	Adj	---	Focus	1,500	---	---	---	---	---

GENERAL:

Qualification - Required

(B) denotes changes

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METHOD	REQUIREMENT OR TEST	NOTES	CONDITIONS	SYMBOL	LIMITS		UNIT
					MINIMUM	MAXIMUM	
	Qualification inspection						
(B) 1216	Base material insulating quality	-	Zone 5 (minimum)	---	---	---	---
1141	Pressure (implosion)	-		---	---	---	---
5111	Vibration	-		Width	---	1.0	mm
1331	Direct-interelectrode capacitance	-	g1 to all	Cg1	---	11.0	pF
			D1 to D2	C1D2	---	4.0	pF
			D3 to D4	C3D4	---	4.0	pF
			D1 to all	CD1	---	15.0	pF
			D2 to all	CD2	---	11.0	pF
			D3 to all	CD3	---	10.0	pF
			D4 to all	CD4	---	11.0	pF
5248	Deflection-factor uniformity	-		---	---	---	---
5246	Focusing voltage, zero-bias	-		Eb1	240	450	Vdc
5101	Neck and bulb alinement (electrostatic types)	-		Diam	---	1.87	Inches
(B) 5216	Cathode illumination	-		---	---	---	---
	Quality conformance inspection, part 1						
5106	Bulb, screen, and faceplate quality	-		---	---	---	---
5201	Voltage breakdown	-		---	---	---	---
5201	Voltage breakdown (electrostatic types)	-		---	---	---	---
5206	Gas "cross"	1	Light - 4 ftL	---	---	---	---
5221	Light output	1		Light	4.0	---	ftL
5223	Modulation	-	Light - 4 ftL	ΔEc1	---	45	Vdc
5231	Spot displacement (leakage)	-		Displ	---	7.0	mm
5231	Spot position (electro-static deflection)	-		---	---	15.0	mm
5241	Grid cutoff voltage	-		Ec1	---	-67.5	Vdc
5251	Grid No. 1 leakage current	-		---	---	---	---
5251	Anode No. 2 leakage current	-		---	---	---	---

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METHOD	REQUIREMENT OR TEST	NOTES	CONDITIONS	SYMBOL	LIMITS		UNIT
					MINIMUM	MAXIMUM	
	Quality conformance inspection, part 2						
1301	Heater current	-		If	540	660	mA
5201	Electrode current (anode No. 1)	-	Ec1 = 0	Ib1	-15	10	μAdc
5201	Electrode current (cathode)	-	Light = 4 ftL	Ik	---	1,000	μAdc
(B) 5101	Base alinement (electrostatic types)	-	+3D4: pin No. 1	---	---	---	---
5101	Angle between traces	-		Angle	87	93	Degrees
5101	Neck and base alinement (electrostatic types)	-		---	---	---	---
(B) 5216	Stray light emission (conventional types)	-		---	---	---	---
5226	Line width A (electrostatic types)	-	Light = 4 ftL	Width	---	0.4	mm
5226	Line width B (electrostatic types)	-	Light = 4 ftL	Width	---	0.4	mm
---	Focusing voltage	-	Light = 4 ftL	Eb1	240	450	Vdc
5248	Deflection factor	-	1D2	DF	75	102	Vdc/in.
5248	Deflection factor	-	3D4	DF	57	78	Vdc/in.
5251	Heater-cathode leakage current	-		---	---	---	---
(B) 1101	Secureness of base, cap, or insert	-		---	---	---	---
(B) 1111	Base pin solder depth	-		---	---	---	---
(B) 1105	Permanence of marking	-		---	---	---	---
	Quality conformance inspection, part 3						
---	Life test	3	Group C; Light = 4 ftL; Eb2 = 2,750 Vdc; t = 500 hours (min)	---	---	---	---
---	Life-test end points:						
	Line width A	-	Light = 3.4 ftL	Width	---	0.4	mm
	Line width B	-	Light = 3.4 ftL	Width	---	0.4	mm
	Modulation	-	Light = 3.4 ftL	ΔEc1	---	45	Vdc

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NOTES:

1. This test to be performed at the conclusion of the holding period.
2. This value is recommended only for low-velocity deflecting and low-ambient light levels.
3. During life-test operation the heater voltage shall be regularly interrrupted on a 13 minutes "on", 7 minutes "off" cycle. Life test minimum duration shall be the total of "on" plus "off" hours.

Custodians:
Army - EL
Navy - EC
Air Force - 80

Review activities:
Air Force - 11, 17
DSA - ES

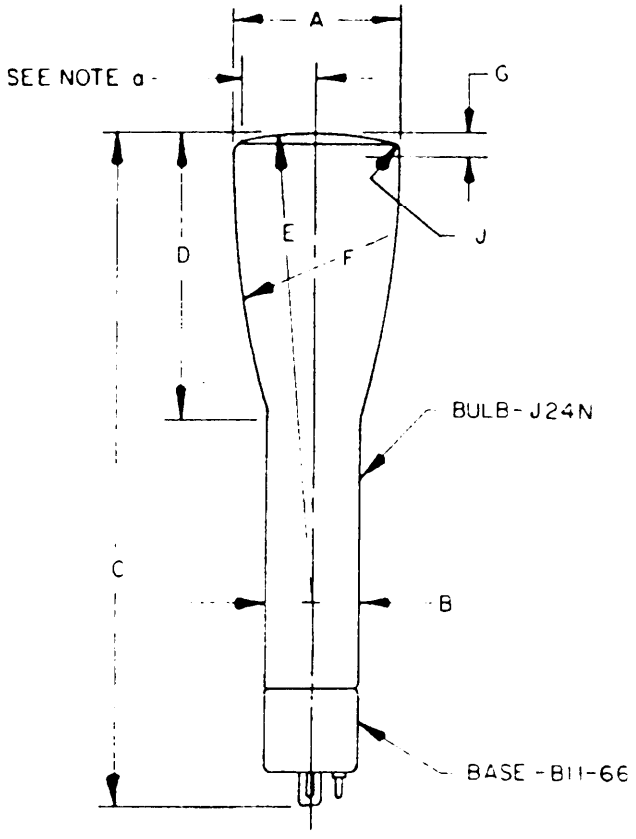
User activities:
Army - WC
Navy - AS, OS, MC, CG, SH

Preparing activity: Army - EL

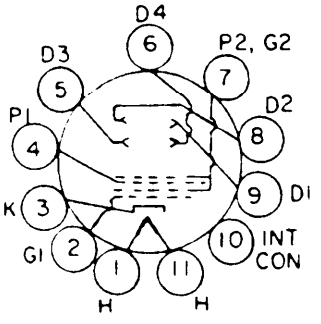
Agent: DSA - ES

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Ltr	Dimensions in inches with metric equivalents (mm) in parentheses	
	Minimum	Maximum
Qualification inspection		
	Base: B11-66	
Quality conformance inspection, part 2		
A	2.937 (74.60)	3.063 (77.80)
B	1.560 (39.62)	1.680 (42.67)
C	11.250(285.75)	11.750(298.45)
Reference dimensions (see note b)		
D	4.875(123.83)	
E	8.000(203.20)Rad.	
F	15.000(381.00)Rad.	
G	.350 (8.89)	
J	.250 (6.35)Rad.	



BASING DIAGRAM

NOTES:
a. The minimum useful screen radius shall not be less than 1.375(34.93 mm).
b. These dimensions are for information only and are not required for inspection purposes.

FIGURE 1. Outline drawing of electron tube type 3KP1.

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION MIL-E-1/501B ELECTRON TUBE, CATHODE RAY TYPE 3KP1

ORGANIZATION

CITY AND STATE | CONTRACT NUMBER

MATERIAL PROCURED UNDER A
☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)* | DATE

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ESC-FM 1068-68

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